

Technical Document #446: Cybersecurity - Comprehensive Overview

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Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Encryption converts plaintext into ciphertext to protect data confidentiality.
- Zero trust architecture assumes no implicit trust and verifies every request.
- Incident response plans define how organizations respond to security incidents.